

Pre-screening

Participant	015
	Yes No
Do you or have you had a history of migraine headaches?	☐ ☒
Do you or have you had a history of claustrophobia?	☐ ☒
Do you or have you had a history of frequent or severe motion sickness?	☐ ☒
Do you or have you had a history of any health problems (e.g., seizures, diabetes, heart problems, vertigo) that affect your ability to drive?	☐ ☒
If you are a female, are you or is there a possibility that you might be pregnant?	☐ ☒

Demographics

1. What is your race/ethnicity?	☒ White / Caucasian
2. What is your gender?	☒ Female
3. How old are you?	27
4. What is the highest level of education you have completed?	☒ Some college/Associates/Technical school training
5. Which option best describes your current home?	☒ I have no permanent residence.
6. How do you pay for healthcare?	☒ Government funding (Medicaid, Medicare, Ryan White, VA)
8. Which best corresponds to your current work situation?	☒ Not working or looking for work.
9. What is your combined family income for the past 12 months (from all sources)	☒ Do not know.

Timeline Followback

Visit 1

Visit 1 Date

06-10-2025

Have you used opioids or any other substances during the assessment period?

☒ Yes

Visit 2

Visit 2 Date

07-01-2025

Have you used opioids or any other substances during the assessment period?

☒ Yes

Visit 3

Visit 3 Date

07-21-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Visit 4

Visit 4 Date

08-04-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Visit 5

Visit 5 Date

08-12-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Phone

Participant # _____

15

System Usability Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently	☐	☐	☒	☐	☐
2. I found the system unnecessarily complex	☒	☐	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☒	☐	☐
6. I thought there was too much inconsistency in this system.	☐	☒	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☒
8. I found the system very cumbersome to use.	☒	☐	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☐	☒
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this app experience.	☐	☐	☒	☐	☐
2. The time I spent in this app just slipped away.	☐	☐	☐	☒	☐
3. I was absorbed in the app task.	☐	☐	☒	☐	☐
4. I felt frustrated while using the app.	☒	☐	☐	☐	☐

5. I found this app to be confusing.	☒	☐	☐	☐	☐
6. Using this app was taxing.	☒	☐	☐	☐	☐
7. The app is attractive.	☐	☐	☐	☒	☐
8. The app is aesthetically appealing.	☐	☐	☐	☒	☐
9. The app appealed to my senses.	☐	☐	☐	☒	☐
10. Using this app was worthwhile.	☐	☐	☒	☐	☐
11. The therapy experience was rewarding.	☐	☐	☒	☐	☐
12. I felt interested in the therapy tasks.	☐	☐	☐	☒	☐

iPAL Application and Intervention Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☒	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☐	☒
3. The videos were videos easy to understand.	☐	☐	☐	☐	☒
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☐	☒

AR

Participant ID _____

15

- | | |
|---|---|
| 1. Which option describes your current home? | ☒ I live with family |
| 2. How do you pay for healthcare? | ☒ Government Funding (Medicaid, Medicare, Ryan White, VA) |
| 3. Which best corresponds to your current work situation? | ☒ Not working or looking for work. |

iGROUP Presence Questionnaire

- | | |
|--|--|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Somewhat Agree |
| 2. I felt that the virtual world surround me. | ☒ Neither Agree nor Disagree |
| 3. I felt like I was perceiving pictures. | ☒ Neither Agree nor Disagree |
| 4. I did not feel present in the virtual space. | ☒ Somewhat Agree |
| 5. I had a sense of acting in the virtual space, rather than operating something form the outside. | ☒ Somewhat Disagree |
| 6. I felt present in the virtual space | ☒ Neither Agree nor Disagree |
| 7. How aware were you of the real world surrounding you while navigating in the virtual world? | ☒ Neither aware nor unaware |
| 8. I was not aware of my real environment. | ☒ Somewhat Disagree |
| 9. I still paid attention to my real environment. | ☒ Somewhat Agree |
| 10. I was completely captivated by the virtual world. | ☒ Somewhat Agree |
| 11 How real did the virtual world seem to you? | ☒ neither real nor unreal |
| 12. How much did your experience in the virtual environment seem consistent with your real-world experience? | ☒ Neither Agree nor Disagree |
| 13. How real did the virtual world seem to you? | ☒ Somewhat unreal |
| 14. The virtual world seemed more realistic than the real world. | ☒ Somewhat Disagree |

System Usability Scale AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☐	☒	☐	☐	☐
2. I found the system unnecessarily complex.	☐	☐	☒	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☒	☐
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☒	☐	☐	☐
6. I thought there was too much inconsistency in this system.	☐	☒	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☒	☐
8. I found the system very cumbersome to use.	☐	☐	☒	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☒	☐
10. I needed to learn a lot of things before I could get going with this system.	☐	☒	☐	☐	☐

User Engagement Scale AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this augmented reality experience	☐	☒	☐	☐	☐
2. The time I spent in augmented reality just slipped away.	☐	☐	☐	☒	☐
3. I was absorbed in the therapy task.	☐	☐	☒	☐	☐
4. I felt frustrated while using the headset.	☐	☒	☐	☐	☐
5. I found this system to be confusing.	☐	☒	☐	☐	☐
6. Using this system was taxing.	☐	☒	☐	☐	☐
7. The virtual environment is attractive	☐	☒	☐	☐	☐

8. The augmented reality is aesthetically appealing.	☐	☒	☐	☐	☐
9. The augmented environment appealed to my senses.	☐	☐	☐	☒	☐
10. Using this system was worthwhile.	☐	☒	☐	☐	☐
11. The therapy experience was rewarding.	☐	☐	☒	☐	☐
12. I felt interested in the therapy tasks.	☐	☐	☒	☐	☐

iPAL Application and Intervention Evaluation AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☒	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☒	☐	☐
3. The videos were clear and easy to understand.	☐	☐	☐	☒	☐
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☒	☐

VR

Participant ID _____ 015

- | | |
|---|---|
| 1. Which option describes your current home? | ☒ I live with family |
| 2. How do you pay for healthcare? | ☒ Government Funding (Medicaid, Medicare, Ryan White, VA) |
| 3. Which best corresponds to your current work situation? | ☒ Not working or looking for work. |

iGroup Presence Scale

- | | |
|--|---|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Completely Agree |
| 2. I felt that the virtual world surrounded me. | ☒ Completely Agree |
| 3. I felt like I was perceiving pictures. | ☒ Completely Agree |
| 4. I did not feel present in the virtual space. | ☒ Completely Disagree |
| 5. I had a sense of acting in the virtual space, rather than operating something from the outside. | ☒ Completely Agree |
| 6. I felt present in the virtual space. | ☒ Completely Agree |
| 7. How aware were you of the real world surrounding you while navigating in the virtual world? | ☒ Somewhat unaware |
| 8. I was not aware of my real environment. | ☒ Somewhat Agree |
| 9. I still paid attention to my real environment. | ☒ Completely Disagree |
| 10. I was completely captivated by the virtual world. | ☒ Completely Agree |
| 11. How real did the virtual world seem to you? | ☒ Very real |
| 12. How much did your experience in the virtual environment seem consistent with your real-world experience? | ☒ Somewhat Disagree |
| 13. How real did the virtual world seem to you? | ☒ Very real |
| 14. The virtual world seemed more realistic than the real world. | ☒ Somewhat Agree |

System Usability Scale VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☐	☐	☐	☐	☒
2. I found the system unnecessarily complex.	☐	☒	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☐	☒	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☐	☒
6. I thought there was too much inconsistency in this system.	☐	☒	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☒	☐	☐	☐
8. I found the system very cumbersome to use.	☐	☒	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☐	☒
10. I needed to learn a lot of things before I could get going with this system.	☐	☒	☐	☐	☐

User Engagement Scale VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this virtual experience	☐	☐	☐	☒	☐
2. The time I spent in the virtual world just slipped away.	☐	☐	☐	☐	☒
3. I was absorbed in the therapy task.	☐	☐	☐	☐	☒
4. I felt frustrated while using the headset.	☐	☒	☐	☐	☐
5. I found this system to be confusing.	☐	☒	☐	☐	☐
6. Using this system was taxing.	☐	☒	☐	☐	☐
7. The virtual environment is attractive	☐	☐	☐	☐	☒

8. The virtual environment is aesthetically appealing.	☐	☐	☐	☐	☒
9. The virtual environment appealed to my senses.	☐	☐	☐	☐	☒
10. Using this system was worthwhile.	☐	☐	☐	☐	☒
11. The therapy experience was rewarding.	☐	☐	☐	☐	☒
12. I felt interested in the therapy tasks.	☐	☐	☐	☐	☒

iPAL Application and Intervention Evaluation VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☐	☒
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☐	☒
3. The videos were clear and easy to understand.	☐	☐	☐	☐	☒

MR

Participant ID _____

015

- | | |
|---|---|
| 1. Which option describes your current home? | ☒ I live with family |
| 2. How do you pay for healthcare? | ☒ Government Funding (Medicaid, Medicare, Ryan White, VA) |
| 3. Which best corresponds to your current work situation? | ☒ Not working or looking for work. |

iGROUP Presence Questionnaire MR

- | | |
|--|--|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Completely Agree |
| 2. I felt that the virtual world surrounded me. | ☒ Completely Agree |
| 3. I felt like I was perceiving pictures. | ☒ Completely Agree |
| 4. I did not feel present in the virtual space. | ☒ Completely disagree |
| 5. I had a sense of acting in the virtual space, rather than operating something from the outside. | ☒ Completely Agree |
| 6. I felt present in the virtual space. | ☒ Completely Agree |
| 7. How aware were you of the real world surround you while navigating in the virtual world? | ☒ Somewhat unaware |
| 8. I was not aware of my real environment. | ☒ Somewhat Agree |
| 9. I still paid attention to my real environment. | ☒ Neither Agree nor Disagree |
| 10. I was completely captivated by the virtual world. | ☒ Somewhat Agree |
| 11. How real did the virtual world seem to you? | ☒ Very real |
| 12. How much your experience in the virtual environment seem consistent with your real-world experience? | ☒ Somewhat disagree |
| 13. How real did the virtual world seem to you? | ☒ Very real |
| 14. The virtual world seemed more realistic than the real world. | ☒ Completely Agree |

System Usability Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☐	☐	☐	☐	☒
2. I found the system unnecessarily complex.	☐	☒	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☐	☒
6. I thought there was too much inconsistency in this system.	☒	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☒	☐
8. I found the system very cumbersome to use.	☐	☐	☐	☐	☒
9. I felt very confident using the system.	☐	☐	☐	☐	☒
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this mixed-reality experience.	☐	☐	☐	☒	☐
2. The time I spent in mixed reality just slipped away.	☐	☐	☐	☐	☒
3. I was absorbed in the therapy task.	☐	☐	☐	☐	☒
4. I felt frustrated while using the headset.	☒	☐	☐	☐	☐
5. I found this system to be confusing.	☒	☐	☐	☐	☐
6. Using this system was taxing.	☒	☐	☐	☐	☐
7. The virtual environment is attractive	☐	☐	☐	☐	☒

8. The mixed-reality is aesthetically appealing.	☐	☐	☐	☐	☒
9. The blended environment appealed to my senses.	☐	☐	☐	☐	☒
10. Using this system was worthwhile.	☐	☐	☐	☐	☒
11. The therapy experience was rewarding.	☐	☐	☐	☐	☒
12. I felt interested in the therapy tasks.	☐	☐	☐	☐	☒

iPAL Application and Intervention Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☒	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☒	☐
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☐	☒

Exit Interview

1. Which option describes your current home?	☒ I live with family
2. How do you pay for healthcare?	☒ Government Funding (Medicaid, Medicare, Ryan White, VA)
3. Which best corresponds to your current work situation?	☒ Not working or looking for work.
1. You experienced Cognitive Behavioral Therapy and Heart Rate Variance interventions in virtual, augmented, and mixed realities. How would you describe your experiences with each of those headsets?	didn't care for AR much, felt like paid attention to external environment too much to actually pay attention to intervention. felt like MR was okay and the easiest to use of the three. VR was the favorite because it was the most immersive and gave illusion that participant was present.
2. Do you think these types of technologies have a place in OUD treatment and management? If YES, how? If NO, why not?	yes - feels like it would be significantly better to use these interventions for individuals as opposed to the standard care with a person, varies on person/patient
3. How would you describe the iPAL application, in terms of a cravings management tool?	feels like it was a good intervention with good questions asked and not too many or too few questions. allows the individual using the app to acknowledge what and how different factors are affecting them.
4. How would you describe the interventions (CBT, HRV)? Were the interventions appropriate for the intensity level and triggers you reported?	CBT - only enjoyed in VR intervention. felt like the videos were too short, but felt like they were appropriate for the intensity level and triggers reported. HRV - enjoyed a lot and felt calm. feels like the intervention would be nice if it were a bit longer and believes the interventions were appropriate for the intensity level and triggers reported.
5. Do you think the iPAL app, or similar digital interventions delivered via phone should be integrated into OUD treatment plans? If YES, how and in what capacity or role? if NO, why not?	yes - could be either used in therapy or at home as needed for triggers or even on a regular basis, depending on the patient and their stability.
6. If iPAL or a similar app was available, would you download the app and incorporate it into your OUD treatment and management plan?	yes - could envision self utilizing app twice per day